Detect, Isolate, Recover: Rotor-Fault Tolerance for a 5-inch Quadrotor (Simulation Study)

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Code and data: `rotor-fault-recovery` (MIT license)

***Abstract*—We study rotor-fault tolerance for a simulated 5-inch FPV-class quadrotor (0.64 kg, thrust-to-weight 3.6) with sensor noise, a 10 ms motion-capture latency, parameter uncertainty and light turbulence. An online estimator of the four rotor effectiveness factors (filtered rigid-body regression, Kalman filter) feeds a bank of CUSUM tests for detection and isolation; a supervisor then either re-allocates thrust with the estimated effectiveness (partial loss) or, when a rotor can no longer carry its share of the weight, switches to a yaw-relaxed controller in the spirit of Mueller and D'Andrea (2014) that spins the vehicle about a tilted primary axis (gain-scheduled LQR on the reduced attitude). In 200 Monte-Carlo runs with 20-90 % loss of effectiveness, all faults were detected (median delay 30 ms, 95th percentile 58 ms) and all were isolated to the correct rotor, with a median magnitude error of 1.5 percentage points; no false alarm occurred in 32 min of fault-free flight. With a 50 % loss in hover the median peak deviation drops from 28 cm (passive) to 4.3 cm. After a complete rotor loss in hover the passive controller lost the vehicle in 32/32 runs (median 0.30 s), while the proposed pipeline recovered 32/32 with a median peak deviation of 38 cm and a spin of 23 rad/s; it also recovered 24/24 complete losses during a 3.3 m/s figure-eight. These are simulation results with assumed, not identified, parameters; in the robustness sweeps complete-loss recovery started to fail at an actuator delay of 24 ms (2 ms assumed), a motor time constant of 80 ms (25 ms assumed) and 3x the default turbulence. In 110 PX4 software-in-the-loop flights of a Gazebo x500, replayed offline with parameters read from the simulator's model files, the same diagnoser detected and isolated all 40 injected partial losses (19-48 %, median delay 120 ms) and all 16 complete losses (median 132 ms), with no false alarm in 22 min of fault-free flight; PX4's own failure detector flagged none of the partial losses and the complete losses only after 664 ms, once the vehicle had flipped.**

I. Introduction

A quadrotor has as many actuators as controlled degrees of freedom, so a damaged propeller, a prop strike or a failed motor channel cannot be compensated by redundancy. Small FPV-class vehicles meet such faults routinely, and their flight software usually treats them only as a reason to disarm. Fault-tolerant control (FTC) of multirotors is a mature topic: Zhang and Jiang [1] review reconfigurable FTC in general, model-based actuator fault diagnosis and accommodation for quadrotors was shown by Avram et al. [2], and the complete loss of a rotor has been handled by giving up yaw: Mueller and D'Andrea derived "relaxed hover" solutions in which the vehicle spins about a body-fixed primary axis and controlled the reduced attitude with LQR [3], [4], Lanzon et al. designed a controller for the same situation [5], and Sun et al. used incremental nonlinear dynamic inversion (INDI) [6] to fly a quadrotor with two opposing rotors lost at high speed [7].

This report puts the pieces together for one specific vehicle class, a 5-inch FPV quadrotor with a thrust-to-weight ratio of 3.6, and asks three questions: how fast and how reliably can a loss of effectiveness (LOE) be detected, isolated and quantified without RPM telemetry; how much does acting on that estimate help compared with a passive controller and with an oracle; and where does yaw-relaxed recovery from a complete loss break. The contribution is not a new control law. It is an end-to-end, reproducible pipeline evaluated under sensor noise, motion-capture latency, parameter spread and turbulence, with its failure limits reported.

II. Model

Vehicle. Position p and velocity v in a z-up world frame, unit quaternion attitude q (rotation R), body rate ω and rotor speeds Ω_i:

$$m\dot{v} = -mge_3 + RTe_3 - RDR^\top v - c\|v\|v + F_d, \quad (1)$$

$$J\dot{\omega} = \tau(f, \dot{\Omega}) - \omega \times (J\omega + h_r) - K_\omega \omega + \tau_d, \quad (2)$$

$$\dot{\Omega}_i = (\Omega_{\mathrm{cmd},i} - \Omega_i)/\tau_m, \quad f_i = \eta_i k_f \Omega_i^2, \quad (3)$$

with rotor angular momentum h_r, yaw reaction torque κf_i per rotor plus the reaction of accelerating rotors, and turbulence F_d, τ_d (independent Ornstein–Uhlenbeck processes, 0.15 N and 4×10^{-3} N m, 0.3 s correlation). $\eta_i \in [0, 1]$ is the effectiveness of rotor i; a fault is a step in one η_i. Mass (0.638 kg) and inertia ($\mathrm{diag}(1.36, 2.12, 3.13) \times 10^{-3}$ kg m^2) come from a lumped mass budget; arm length 0.11 m, 5.6 N maximum rotor thrust, $\kappa = 0.015$ m, $\tau_m = 25$ ms, rotational damping $(2.5, 2.5, 2.0) \times 10^{-3}$ N m s. The true vehicle of each run is drawn around these values (mass $\pm 3\%$, inertia $\pm 5\%$, k_f $\pm 3\%$ per rotor, τ_m $\pm 10\%$, damping $\pm 20\%$); the flight software always uses the nominal model. Integration is RK4 at 1 kHz; estimation, FDI and control run at 500 Hz with 2 ms actuator transport delay.

Sensors. Gyro with 0.03 rad/s white noise per sample, random-walk bias and $\pm 2000°$/s range; accelerometer with 0.2 m/s^2 noise and bias; motion capture at 100 Hz with 1 mm / 0.3° noise and 10 ms latency. A complementary attitude filter and a position/velocity Kalman filter fuse them, using time-stamped history for the delayed motion-capture samples. Controllers and the diagnoser never see the true state.

III. Nominal control and allocation

The nominal controller is the geometric tracking controller on SE(3) of Lee et al. [8] (z-up form), with integral action on position and attitude, tilt limits and drag feed-forward; its SO(3) attitude error has no singularities and behaves well for the large errors that follow a fault. Allocation solves $[T, \tau]^\top = B\,\mathrm{diag}(\eta) f$ under $f_{\min} \le f \le f_{\max}$ in closed form with priorities: the roll/pitch torque is kept (scaled only if infeasible), the collective is shifted by the smallest feasible amount, and the yaw torque is scaled last.

IV. Fault detection, isolation and estimation

Regression. The rotational dynamics and the body-z force balance are linear in η:

$$y \triangleq \begin{bmatrix} m a_z + D_z v_z + c\|v\| v_z \\ J\dot{\omega} + \omega \times (J\omega + h_r) + K_\omega \omega - \tau_{\hat{\Omega}} \end{bmatrix} = B\,\mathrm{diag}(\hat{f})\,\eta \triangleq \Phi\eta, \tag{4}$$

where $\hat{f}$ are thrusts predicted from the commands through the nominal motor model and the known delay. $\dot{\omega}$ comes from a 15 Hz second-order state-variable filter of the gyro, and every other signal of the regression passes through the same filter.

Estimation. A Kalman filter with a random-walk model on η (equivalently RLS with forgetting) tracks $\hat{\eta}$, projected onto $[0, 1.1]$; rotors commanded near idle are frozen. Static mismatch (3 % k_f spread, mass error) is indistinguishable from a small LOE, so the estimate is averaged over the initial 2.5 s hover into a healthy baseline η_{ref}.

Detection and isolation. Each residual row is normalised by $\sigma_k = \sqrt{\sigma_0^2 + u_k^2}$, a calibrated floor plus the residual that the assumed inertia (5 %) and motor-lag (10 %) uncertainty can produce at that instant. A loss δ on rotor j shifts $r = (y - \Phi\eta_{\mathrm{ref}})/\sigma$ by δd_j with the known signature $d_j = -\Phi_{:,j}\eta_{\mathrm{ref},j}/\sigma$. With $t_j = d_j^\top r / \|d_j\|$, a bank of one-sided CUSUM tests [9], [10]

$$g_{kj} \leftarrow \max\big(0,\ g_{kj} + \mu_{kj}(t_j - \mu_{kj}/2)\big), \quad \mu_{kj} = \delta_k \|d_j\|, \tag{5}$$

for $\delta_k \in \{0.15, 0.3, 0.6\}$ approximates the GLR test for an unknown fault size (cf. Lorden [11]); an alarm is raised when any $g_{kj} > h_k$ and rotor $\arg\max_j g_{kj}$ of the firing test is isolated. Because the residual is coloured and the model inexact, σ_0 and h_k are calibrated on 60 fault-free flights with seeds disjoint from all evaluation runs: h_k is twice the largest statistic observed, floored at the evidence a design-size fault accumulates in 20 ms. The aggressive waypoint manoeuvre is excluded from calibration.

V. Fault-tolerant control

Partial loss. After an alarm, allocation uses $B\,\mathrm{diag}(\hat{\eta})$.

Complete loss. If $\hat{\eta}_j < 0.35$ for 30 ms (below ≈ 0.29 the rotor cannot carry a quarter of the weight), rotor j is idled and a yaw-relaxed controller in the spirit of [3], [4] takes over. The relaxed-hover equilibrium is computed numerically: three thrusts and a constant body rate $\bar{\omega} = \bar{r}\,\bar{n}/\bar{n}_z$ along a body-fixed axis $\bar{n}$ such that all torques balance (including rotor gyroscopics and damping) and $f_T \bar{n}_z = mg$. The solutions form a two-parameter family; I fix the tilt of $\bar{n}$ from body z at 15° and choose its azimuth to minimise the spin (Fig. 6), giving 24 rad/s, 69 % of the gyro range. The reduced dynamics – p, q, the two components of the axis error in the plane orthogonal to $\bar{n}$ (with $\dot{n} = -\omega \times n$ for the desired axis expressed in the body frame) and three motor states – are linearised numerically about quasi-equilibria with the yaw rate frozen at 14 values between 0 and $1.3\bar{r}$; a discrete LQR is computed at each point and scheduled on the measured yaw rate, as is the gyroscopic torque feed-forward. An outer position loop commands the direction of $\bar{n}$ and $f_T = m\|a_{\mathrm{cmd}}\|/\bar{n}_z$.

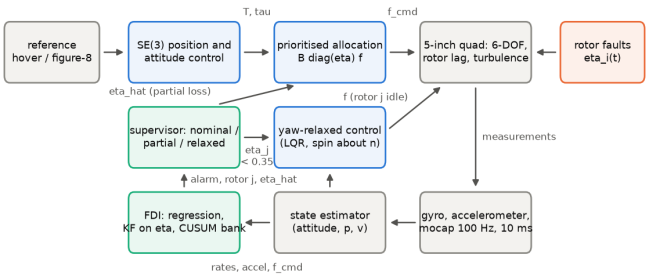


Fig. 1. Pipeline. The supervisor switches between nominal, partial-loss (re-allocation with $\hat{\eta}$) and yaw-relaxed control.

Baselines. *Passive*: nominal controller, faults ignored. *Oracle*: the true η and onset are given to the supervisor. *Estimated*: the proposed pipeline. For a given seed, all three see the same vehicle, noise and turbulence.

VI. Experiments

E1: hover, step LOE 0–90 % on rotor 1 at $t = 2$ s, 8 seeds per point. E2: 200 faulty runs (random rotor, LOE $\sim \mathcal{U}[0.2, 0.9]$, onset $\sim \mathcal{U}[2, 5]$ s, hover or a figure-eight at up to 3.3 m/s) and 100 fault-free runs of 12 s (hover, figure-eight, aggressive waypoint steps), plus a gyro-noise sweep (0.5–8×, re-calibrated per level). E3: figure-eight with 50 % LOE at $t = 10$ s. E4/E5: complete loss of each rotor in hover / on the figure-eight. E6: robustness of E4 to the true actuator delay, motor time constant, damping, inertia and turbulence. E7: FDI ablation. A run is *lost* on ground contact, tilt $> 90°$ or position error > 3 m.

VII. Results

Partial loss (E1, E3). With 50 % LOE in hover the median peak deviation drops from 28 cm (passive) to 4.3 cm, which is the turbulence floor (Fig. 2, Table I). The passive controller recovers partial losses eventually through its integrators but loses 3/8 runs at 70 % and 8/8 at 90 %; the active pipelines lose none. Two unexpected results: at 80 % the passive vehicle survives in an uncontrolled but stable yaw spin, and at 70–80 % the estimated pipeline beats the oracle because the oracle idles the weakened rotor immediately. On the figure-eight (Table II) the median peak deviation after a 50 % loss is 25 cm passive and 6.8 cm estimated.

Diagnosis (E2, E7). All 200 faults were detected (median delay 30 ms, 95th percentile 58 ms; about 51 ms for 20–35 %

TABLE I

E1: HOVER WITH A STEP LOSS OF EFFECTIVENESS ON ROTOR 1. HOVER, FAULT AT T = 2 S. EACH CELL: MEDIAN MAX DEVIATION / MEDIAN SETTLED ERROR (CM) OVER NON-LOST RUNS, (LOST/TOTAL).

LOE rotor 1	Passive	Oracle	Estimated (proposed)
20 %	7.6 / 2.7 (0/8)	4.3 / 2.5 (0/8)	4.3 / 2.2 (0/8)
40 %	18.1 / 3.3 (0/8)	4.3 / 2.5 (0/8)	4.3 / 2.2 (0/8)
60 %	46.1 / 5.5 (0/8)	4.3 / 2.5 (0/8)	4.4 / 2.2 (0/8)
70 %	73.5 / 16.1 (3/8)	19.5 / 5.6 (0/8)	12.9 / 5.5 (0/8)
80 %	97.7 / 31.0 (0/8)	19.8 / 5.7 (0/8)	18.7 / 5.5 (0/8)
90 %	– / – (8/8)	20.1 / 5.8 (0/8)	27.2 / 6.2 (0/8)

TABLE II

E3: FIGURE-EIGHT TRACKING WITH A 50 % LOSS. FIGURE-EIGHT (PEAK 3.3 M/S), 50 % LOE ON ROTOR 1 AT T = 10 S; MEDIANS OVER 10 SEEDS, CM.

Baseline	RMSE before	RMSE 0-3 s after	RMSE 3-10 s after	max dev.	lost
Passive	3.5	17.6	8.5	24.7	0/10
Oracle	3.5	3.7	4.0	7.0	0/10
Estimated (proposed)	3.5	3.5	3.9	6.8	0/10

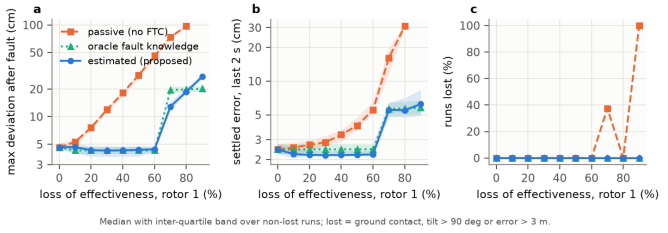


Fig. 2. E1: peak deviation, settled error and loss rate versus LOE magnitude (median and IQR, 8 seeds).

and 24 ms for 80–90 %), 100 % were isolated correctly and the median magnitude error was 1.5 percentage points (Fig. 3, Table III); 0 false alarms occurred in 32 min of fault-free flight, including aggressive waypoint steps that were not used for calibration. Up to 8× gyro noise (14°/s per sample), detection and isolation are essentially unchanged, with re-calibrated thresholds and also with the 1× thresholds (Table IV): in this model turbulence and mismatch, not white gyro noise, set the residual floor. The ablation (Table V) shows that without the healthy baseline static mismatch causes false alarms (1/30 fault-free runs and 1 faulty run before its onset), and that a single CUSUM designed for 15 % roughly doubles the median delay (61 vs. 30 ms). The adaptive residual scale made no measurable difference in these tests (no false alarms without it, 25 ms median delay), so no benefit is claimed for it.

Complete loss (E4, E5). The passive controller lost all 32 hover runs (median 0.3 s after the failure) and all 24 figure-eight runs. The estimated pipeline lost 0/32 and 0/24, switching to yaw-relaxed control a median 62 ms after the failure; the median peak deviation in hover was 38 cm (oracle 22 cm), the settled error 7.6 cm and the spin 23 rad/s (Fig. 4, Table VI).

Limits (E6). Every run recovered over the whole tested range of rotational damping (0.25-2 x nominal) and inertia

TABLE III

E2: FAULT DETECTION AND ISOLATION MONTE CARLO. 200 FAULTY RUNS (HOVER AND FIGURE-EIGHT). FALSE ALARMS: 0 IN 32 MIN OF FAULT-FREE FLIGHT (100 FAULT-FREE RUNS INCL. UNSEEN AGGRESSIVE STEPS, PLUS PRE-ONSET TIME OF THE FAULTY RUNS).

LOE	detected	delay ms: median (p95)	isolation	abs. LOE error, pts: median (p95)
20-35 %	33/33	51 (80)	100.0 %	1.7 (4.6)
35-50 %	44/44	37 (43)	100.0 %	1.0 (4.4)
50-65 %	44/44	30 (35)	100.0 %	1.1 (2.7)
65-80 %	46/46	27 (32)	100.0 %	1.3 (3.7)
80-90 %	33/33	24 (27)	100.0 %	3.8 (5.4)
all 20-90 %	200/200	30 (58)	100.0 %	1.5 (4.8)

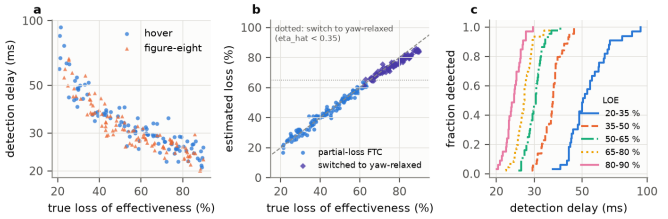


Fig. 3. E2: detection delay versus LOE, estimated versus true LOE, and delay distribution per magnitude bin (200 runs).

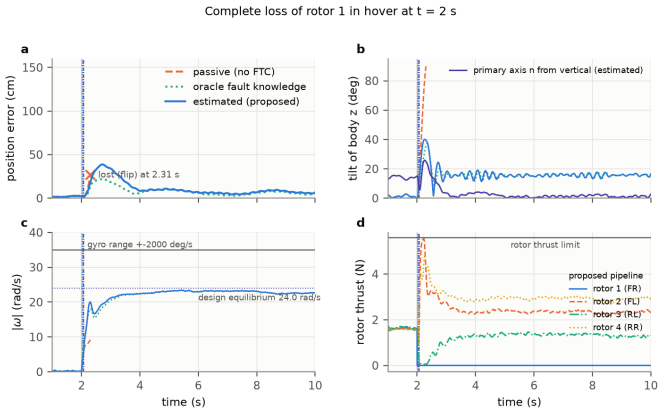


Fig. 4. E4: complete loss of rotor 1 in hover: position error, tilt, spin rate and rotor thrusts.

(0.7-1.4 x nominal). Runs were lost for: actuator delay (24 ms: 5/8 lost; tested 0-24 ms, nominal 2); motor time constant (80 ms: 8/8 lost, 100 ms: 8/8 lost; tested 15-100 ms, nominal 25); turbulence (3 x default: 5/8 lost, 4 x default: 6/8 lost; tested 0-4 x default, nominal 1). The design tolerates the inertia

TABLE IV

E2: SENSITIVITY TO GYRO NOISE. 1X = MAIN EXPERIMENT (200 + 100 RUNS); OTHER LEVELS 40 FAULTY + 20 FAULT-FREE RUNS WITH RESIDUAL FLOOR AND THRESHOLDS RE-CALIBRATED FOR THAT LEVEL; THE SECOND FALSE-ALARM COLUMN RE-FLIES THE 20 FAULT-FREE RUNS WITH THE THRESHOLDS CALIBRATED AT 1X.

gyro noise (deg/s)	detected	delay ms: median (p95)	isolation	abs. LOE error, pts	false alarms	false alarms, 1x thresholds	lost
0.5x (0.9)	40/40	18 (30)	100.0 %	1.1	0	0/20 runs	0
1x (1.7)	200/200	30 (58)	100.0 %	1.5	0	–	0
2x (3.4)	40/40	26 (34)	100.0 %	1.3	0	0/20 runs	0
4x (6.9)	40/40	29 (53)	100.0 %	1.7	0	0/20 runs	0
8x (13.8)	40/40	25 (48)	100.0 %	2.0	0	0/20 runs	0

TABLE V

E7: FDI ABLATION. 60 FAULTY + 30 FAULT-FREE RUNS PER VARIANT (SAME SEEDS), EACH VARIANT CALIBRATED WITH THE SAME PROCEDURE; DETECTED COUNTS ALARMS AFTER ONSET.

FDI variant	detected	delay ms: median (p95)	isolation	abs. LOE error, pts	false-alarm runs (+ before onset)	lost
proposed (all elements)	60/60	30 (64)	100.0 %	1.6	0/30 (+0)	0
no healthy baseline	59/60	28 (63)	100.0 %	1.8	1/30 (+1)	0
no adaptive residual scale	60/60	25 (56)	100.0 %	1.2	0/30 (+0)	0
single CUSUM (15 %)	60/60	61 (99)	100.0 %	1.2	0/30 (+0)	0

and damping errors tested and motors up to 2.4× slower than assumed, but fails at 24 ms actuator delay, 80 ms motor time constant and 3× the default turbulence. All three act during the spin-up transient, where the relaxed equilibrium leaves little margin and the body tilts by about 40° (Fig. 5, Table VII).

VIII. PX4 SITL VALIDATION

To check that the diagnoser is not tied to the simulator it was developed on, it was replayed offline on ULogs of PX4 [12] software-in-the-loop flights: PX4-Autopilot (v1.18.0-beta1, unmodified) with its x500 quadrotor in Gazebo Harmonic [13], run headless and commanded over MAVLink to take off, hover or fly a 5 m/3 m/s orbit, in calm air or PX4's windy world (steady 5.4 m/s), with a fault injected 8–14 s into the manoeuvre. The x500 weighs 2.1 kg and has a thrust-to-weight ratio of 1.7. Its airframe description (mass, inertia, rotor layout, spin directions, motor constants and asymmetric time constants) is generated from PX4's model files; only the command-to-thrust delay (4 ms) is fitted, and the residual floor and thresholds come from 24 fault-free calibration flights with disjoint seeds, by the procedure of Section IV. All filter, Kalman-filter and CUSUM settings are unchanged.

PX4's own `failure motor off` injection (the only motor failure type at this version) had no effect with the default `CA_FAILURE_MODE` = 0, and with mode 1 it logs the stopped motor's command as NaN, which reveals the fault to a log-based diagnoser. The evaluated faults were therefore injected in the simulated ESC: the limits of the bridge's command-to-speed map are scaled at runtime, so that one motor delivers a fraction of the commanded speed while PX4 keeps logging the intended command.

In 110 PX4 SITL flights (24 of them for calibration), the diagnoser detected 40/40 partial losses of 19-48 % (median delay 120 ms, 95th percentile 128 ms), isolated 40/40 to the right motor and estimated the loss with a median error of 0.5 points; it detected 16/16 complete losses after a median 132 ms, all before the vehicle tilted past 90 deg (median 486 ms). No false alarm occurred in 22 min of fault-free flight. PX4's failure detector raised no flag for the partial losses, and for the complete losses its attitude-limit check fired 16/16 times after a median 664 ms, in 16 of them after the tilt had passed 90 deg. Without the adaptive residual scale the median delays drop to 40 ms (partial) and 20 ms (complete), but 4 of 56 faults are then isolated to the wrong motor. After a partial loss PX4 held altitude in 15/16 runs below 35 % loss and in 6/24 runs above. PX4's own failure motor off injection changed nothing in 2/2 flights with the default CA_FAILURE_MODE = 0; with CA_FAILURE_MODE = 1 it stopped the motor and the vehicle flipped in 4/4 (attitude check after a median 660 ms).

The comparison with PX4 (Table VIII, Fig. 7) is not one between two diagnosers: PX4's checks are last-resort triggers (attitude beyond 60° for 0.3 s, altitude loss, off by default, and ESC current against throttle, which cannot fire in SITL because Gazebo's ESC model reports no current). The adaptive residual scale, for which the 5-inch ablation showed no benefit, trades about 0.1 s of delay for correct isolation here: the four isolation errors without it all named the diagonal partner with the same spin direction, within the first tens of milliseconds. The false-alarm result is weak evidence: Gazebo's IMU is almost noise-free and does not vibrate, the calibration statistics never left zero and the thresholds were set by the 20 ms evidence floor. No fault-tolerant control was run in PX4.

IX. DISCUSSION AND LIMITATIONS

The results support the pipeline *in simulation*: diagnosis is fast enough that the partial-loss transient stays at the turbulence floor, and it is fast enough to hand a complete loss to the yaw-relaxed controller within about 60 ms. They do not show that it works on a real 5-inch quadrotor; the PX4 SITL replay shows that the diagnoser carries over to another flight stack and vehicle model, which is still simulation. All

TABLE VI

E4/E5: COMPLETE LOSS OF ONE ROTOR. COMPLETE LOSS OF ONE ROTOR (ALL FOUR ROTORS IN TURN, SEVERAL SEEDS EACH). SPIN = MEAN $|\omega|$ OVER THE LAST 2 S; SWITCH DELAY = ONSET TO YAW-RELAXED CONTROL.

case	baseline	lost	time to loss (s)	max dev. (cm)	settled (cm)	spin (rad/s)	switch delay (ms)
hover	Passive	32/32	0.30	–	–	–	–
hover	Oracle	0/32	–	21.5	6.4	23.2	0
hover	Estimated (proposed)	0/32	–	37.9	7.6	23.2	62
figure-eight	Passive	24/24	0.29	–	–	–	–
figure-eight	Oracle	0/24	–	25.4	16.7	23.6	0
figure-eight	Estimated (proposed)	0/24	–	39.2	18.3	23.6	62

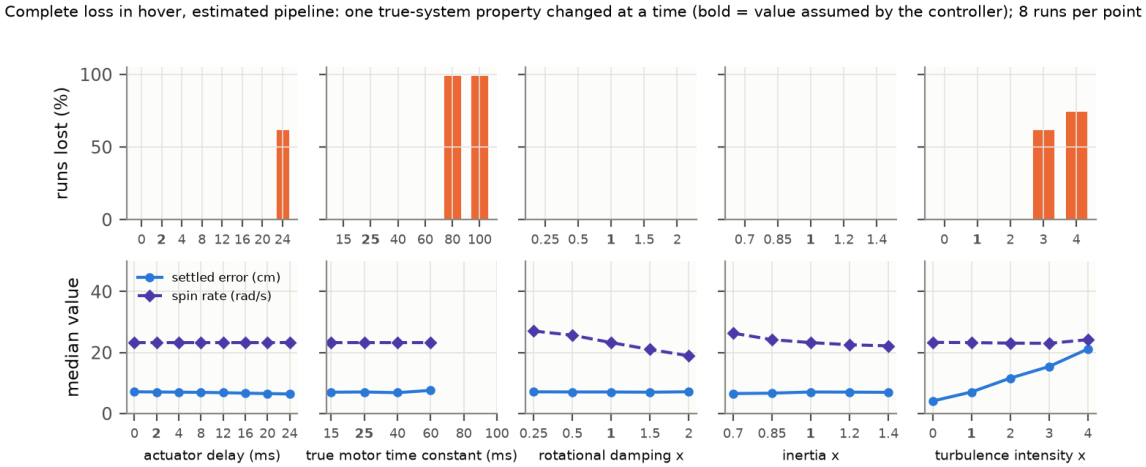


Fig. 5. E6: robustness of complete-loss recovery (estimated pipeline) when one property of the true system departs from the model.

TABLE VII

E6: ROBUSTNESS OF COMPLETE-LOSS RECOVERY. COMPLETE LOSS IN HOVER, ESTIMATED PIPELINE; THE CONTROLLER ALWAYS ASSUMES THE NOMINAL MODEL.

true-system change	runs lost per value
actuator delay (ms)	0: 0/8, 2: 0/8, 4: 0/8, 8: 0/8, 12: 0/8, 16: 0/8, 20: 0/8, 24: 5/8
motor time constant (ms)	15: 0/8, 25: 0/8, 40: 0/8, 60: 0/8, 80: 8/8, 100: 8/8
rotational damping (x)	0.25: 0/8, 0.5: 0/8, 1: 0/8, 1.5: 0/8, 2: 0/8
inertia (x)	0.7: 0/8, 0.85: 0/8, 1: 0/8, 1.2: 0/8, 1.4: 0/8
turbulence (x)	0: 0/8, 1: 0/8, 2: 0/8, 3: 5/8, 4: 6/8

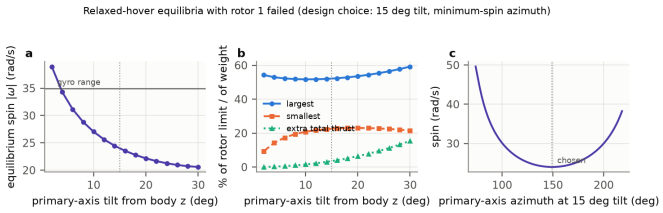


Fig. 6. Relaxed-hover equilibria after the loss of rotor 1 versus primary-axis tilt and azimuth.

parameters are assumed; the rotational damping that sets the spin rate is the least certain, and the aerodynamics of a body spinning at more than 20 rad/s is not modelled. Real vibration is narrow-band and larger than the white noise used here, and the turbulence model is a simple Gauss–Markov process, so the near-perfect detection numbers are an upper bound. Faults are single steps; intermittent or growing damage, multiple faults and sensor faults are not studied, the baseline assumes a fault-free initial hover, and the diagnoser is not valid during the relaxed spin. Motion capture is assumed; GNSS and magnetometer estimation while spinning is harder. Thresholds are calibrated from maxima of a few dozen flights, which is a noisy statistic. The oracle is an upper bound on information, not on the switching policy.

X. CONCLUSION AND NEXT STEPS

A model-based diagnoser (filtered rigid-body regression, Kalman filter, CUSUM bank) combined with re-allocation for partial losses and a gain-scheduled yaw-relaxed controller for complete losses recovered every simulated rotor fault in the main experiments, including complete losses during 3.3 m/s flight; in the robustness sweeps it failed at large actuator delay, very slow motors and strong turbulence. Replayed on PX4 SITL logs of a different vehicle, it detected 40/40 partial and 16/16 complete losses, with 0 false alarms in 22 min of fault-free flight. The next steps are tethered tests on the real 5-inch quadrotor with the same estimator replayed on Betaflight blackbox logs (fault-free flights for the residual floor and the parameters, then a weakened propeller and a throttled ESC; the log pipeline and the protocol are in the repository), the diagnoser running inside PX4, and a comparison with INDI-based FTC [6], [7].

TABLE VIII

PX4 SITL VALIDATION. PX4 V1.18.0-BETA1-322-G18098D3BBE SITL, X500 IN GAZEBO HARMONIC; THE DIAGNOSER IS REPLAYED OFFLINE ON THE ULOGS. PARTIAL: THRUST LOSS OF 19-48 % ON A RANDOM MOTOR, ONSET 8-14 S INTO THE MANOEUVRE; COMPLETE: EACH MOTOR IN EACH CONDITION (THE LOSS ESTIMATE IS CLIPPED AT 100 %). FALSE ALARMS: 0 (AS DESIGNED) AND 0 (NO ADAPTIVE SCALE) IN 22 MIN OF FAULT-FREE FLIGHT (24 HELD-OUT FLIGHTS PLUS THE TIME BEFORE EACH ONSET); CALIBRATION ON 24 OTHER FAULT-FREE FLIGHTS. PX4 AFTER THE FAULT, WITHIN 8 S (6 S FOR COMPLETE LOSSES): HELD ALTITUDE (LESS THAN 1 M LOST), DESCENDED (MORE THAN 1 M LOST, TILT BELOW 90 DEG) OR FLIPPED.

fault	condition	runs	detected	isolated	delay ms: median (p95)	no adaptive scale: delay (p95), isolated	abs. loss error, pts	PX4 failure detector	PX4 after the fault
partial	hover, calm	10	10/10	10/10	120 (126)	40 (50), 10/10	0.1	none	held 4, descended 6
partial	orbit, calm	10	10/10	10/10	120 (128)	42 (56), 10/10	0.9	none	held 8, descended 2
partial	hover, wind	10	10/10	10/10	124 (124)	38 (48), 10/10	0.4	none	held 6, descended 4
partial	orbit, wind	10	10/10	10/10	122 (137)	36 (55), 8/10	2.3	none	held 3, descended 6, flipped 1
partial	all	40	40/40	40/40	120 (128)	40 (56), 38/40	0.5	none	held 21, descended 18, flipped 1
complete	all four	16	16/16	16/16	132 (141)	20 (24), 14/16	0.0	fd_roll 16/16, 664 ms	flipped 16

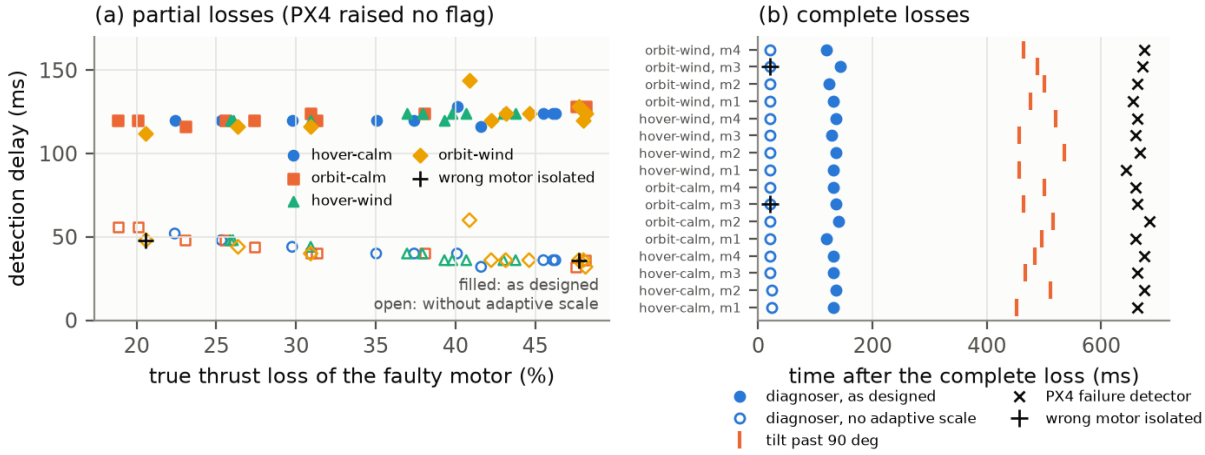


Fig. 7. PX4 SITL: (a) detection delay of the partial losses versus the true thrust loss, as designed (filled) and without the adaptive residual scale (open); (b) complete losses: diagnoser alarm, the instant the tilt passes 90°, and PX4's failure detector.

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